GATE

Computer Science & Information Technology

Computer Networks

Routing

DPP: 1

Q1 Consider the following three statements about link state and distance vector routing protocols for a large network with 500 network nodes and 4000 links.
[S1] The computational overhead in link state protocols is higher than in distance vector protocols.
[S2] A distance vector protocol (with a split-horizon) avoids persistent routing loops but not a link-state protocol.
[S3] After a topology change, a link-state protocol will converge faster than a distance vector protocol.
Which of the following is correct about S1, S2, and S3?
(A) S1, S2, and S3 are all true.
(B) S1, S2, and S3 are all false
(C) S1 and S2 are true, but S3 is false.
(D) S1 and S3 are true, but S2 is false

Q2 Which of the following is TRUE about the interior gateway routing protocols -Routing Information Protocol (RIP) and Open Shortest Path First (OSPF)?
(A) RIP uses distance vector routing, and OSPF uses link state routing
(B) OSPF uses distance vector routing, and RIP uses link state routing
(C) Both RIP and OSPF use link-state routing
(D) Both RIP and OSPF use distance vector routing

Q3 What are all the fields required from the IP header to allow the destination to perform the reassembly of fragments?
(A) Identification (B) MF, Offset
(C) Header length (D) Total length

Q4 Two popular routing algorithms are Distance Vector routing (DVR) and Link State. Routing (LSR) routing. Which of the following is true?
(A) Count-to-infinity is a problem only with DVR, not in LSR.
(B) In LSR, the shortest path algorithm is run only at one node.
(C) In DVR, the shortest path algorithm runs only at one node.
(D) DVR requires a lesser number of messages than LSR.

Q5 Which statements about routing and flooding methods are correct?
(A) In the Routing method, there is a Routing Table that is a disadvantage compared to the flooding method.
(B) In the routing method, reliability is low, and reliability is high in the flooding method.
(C) In the Routing method, There is no duplicate packet, but in the flooding method, there are duplicate packets.
(D) In the Routing method, Traffic is low, but in the flooding method, traffic is high.

Q6 Which properties are/ are False about following the Routing Algorithm?
(A)

Android App | iOS App | PW Website

In Static Routing, everything is set up manually & requires administration, and In Dynamic Routing, everything is done by routers Automatically

(B) Static Routing works when traffic is predictable & network is simple, and Dynamic Routing calculates routes based on received updated network state information.

(C) Static Routing and Dynamic Routing, which adapt to changes depending on traffic and topology.

(D) LSR is a Static routing algorithm, and DVR is the Dynamic routing algorithm

Android App | iOS App | PW Website

GATE

Answer Key

Q1	(D)	Q4	(A, D)
Q2	(A)	Q5	(A, B, C, D)
Q3	(A, B, C, D)	Q6	(C, D)

Android App | iOS App | PW Website

Hints & Solutions

Q1 Text Solution:

S1: The computational overhead in link state protocols is higher than in distance vector protocols. Because LSR is based on global knowledge, whereas DVR is based on local information. (True)

S2: A distance vector protocol with a split-horizon to avoid persistent routing loops is true, but not a link-state protocol is false because link-state protocols do not have a count-to-infinity problem.

S3: As the distance vector protocol has a count-to-infinity problem, it converges slower. (True)

Q2 Text Solution:

RIP Uses Distance Vector Routing, and OSPF uses Link State Routing.

Q3 Text Solution:

All the fields are required to perform Reassembly.

Q4 Text Solution:

A- True - Count-to-infinity is a problem only with DVR, not in LSR.

B- False - In LSR, the shortest path algorithm is run only at one node.

C- False - In DVR, the shortest path algorithm runs only at one node.

D- True - DVR requires fewer messages than LSR.

Q5 Text Solution:

A- TRUE - In the Routing method, the Routing Table is a disadvantage compared to the flooding method.

B- TRUE - In the routing method, reliability is low, and in the flooding method, reliability is high.

C- TRUE - In the Routing method, There is no duplicate packet, but in the flooding method, there are duplicate packets.

D- TRUE - In the Routing method, Traffic is low, but in the flooding method, traffic is high.

Q6 Text Solution:

A- TRUE - Static Routing is set up manually and requires administration, and In Dynamic Routing, everything is done by routers Automatically.

B- TRUE - Static Routing works when traffic is predictable & network is simple, and Dynamic Routing calculates routes based on received updated network state information.

C- FALSE - Static Routing does not change based on network traffic and topology, But Dynamic Routing does change depending on network traffic and topology.

D- FALSE - LSR and DVR both are Dynamic routing algorithm

Android App | iOS App | PW Website